

Assignment for Java Memory Management and Java Collection

Part 1: Java Memory Management (Theory Questions)

1. Explain the different memory areas managed by JVM.
 - Stack Memory
 - Heap Memory
 - Method Area
 - Program Counter (PC) Register
 - Native Method Stack
 2. Differentiate between Stack and Heap memory with examples.
 3. What is Garbage Collection? How does it work in Java?
 4. What are the different types of Garbage Collectors available in Java? Mention any two.
 5. What is the role of the `finalize()` method in Java? Is it recommended to use it? Why/Why not?
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Part 2: Java Collections (Practical Programming Questions)

Collection Hierarchy & List

6. Draw a simple diagram (can be hand-drawn) of Java's Collection hierarchy up to List and its child classes.

ArrayList

7. Write a Java program to:

- Create an `ArrayList` of Strings.
- Add 5 names to the list.
- Print all names using a `for` loop and an **enhanced for loop**.
- Remove the 3rd element.
- Print the updated list.

LinkedList

8. Write a Java program to:

- Create a `LinkedList` of Integers.
- Add numbers `10, 20, 30, 40, 50`.
- Add `5` at the beginning and `60` at the end.
- Remove the element at index 2.
- Print all elements.

Stack

9. Write a Java program to:

- Create a Stack of Integers.
- Push 5 numbers onto the stack.

- Pop 2 elements.
- Print the remaining elements.

10. **Bonus Question:** Explain with an example how Stack follows **LIFO** (Last In First Out) principle.

Vector

11. Write a Java program to:

- Create a Vector of **Double** values.
 - Add 4 decimal numbers.
 - Iterate using an **Enumeration**.
 - Print all values.
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Part 3: Conceptual & Application

12. **Difference-based Questions:**

- ArrayList vs LinkedList
- Stack vs Vector
- ArrayList vs Vector

13. **Real-life Scenario:**

Suppose you are developing a **browser history feature**. Which Java Collection would you use to:

- Store the visited URLs so that you can go **back** and **forward** easily?
- Explain your reasoning.